

Lithography Defect Detection Using Deep Learning

December 2025

1 Assumptions and Solution Approach

1.1 Problem Statement

The objective is to detect defects in lithography/SEM images of semiconductor wafers. The primary challenge is the **domain gap** between available training data (PCB defect images) and the target domain (lithography wafer images). Additionally, we must detect **bright particle contamination defects** that are not present in the original training dataset.

1.2 Key Assumptions

1. **One-Class Detection:** All defect types (open circuits, shorts, mouse bites, pinholes, spurs, excess copper) are treated as a single “defect” class vs. background. This simplifies the detection task and improves generalization.
2. **Domain Adaptation via Simulation:** Physics-based simulation can bridge the gap between PCB training images and lithography target images. The optical lithography process can be approximated using Airy disk point spread functions (PSF) and resist development models.
3. **Synthetic Data Augmentation:** Bright particle defects (contamination) can be synthesized and injected into training images to teach the model to detect this defect type without requiring real annotated examples.
4. **Bounding Box Detection:** Defects are localized using axis-aligned bounding boxes with confidence scores.

1.3 Dataset

We use the **PCB Defect Detection** dataset from Roboflow Universe¹, which contains 1,504 annotated images of PCB defects in COCO format. The dataset includes various defect types (open circuits, shorts, mouse bites, pinholes, spurs, and spurious copper) which we consolidate into a single defect class for training.

1.4 Model Architecture

We use **Faster R-CNN** with a **ResNet-50-FPN** backbone, pretrained on ImageNet. The box predictor head is replaced with a 2-class classifier (background + defect). This architecture provides:

- Multi-scale feature extraction via Feature Pyramid Network (FPN)
- Region proposal network (RPN) for candidate generation
- Robust pretrained features for transfer learning

¹<https://universe.roboflow.com/pevi/pcb-defect-detection-upbty>

1.5 Two-Stage Training Pipeline

Stage 1: Baseline Training - Train on original PCB defect images (RGB, 640×640 pixels) with standard augmentation. This establishes baseline defect detection capabilities on clean, high-contrast images.

Stage 2: Domain Adaptation Fine-tuning - Fine-tune the Stage 1 model with a physics-based augmentation pipeline:

1. **SEM-Style Conversion:** Grayscale conversion with inverted contrast mapping, Gaussian noise, grain texture, and vignetting to simulate scanning electron microscope imagery.
2. **Optical Lithography Simulation:** Convolve with Airy disk PSF ($\lambda = 193\text{nm}$, $\text{NA} = 1.2$), apply sigmoid resist development model, and add line edge roughness (LER) noise.
3. **Synthetic Particle Injection:** Add bright circular particles (10–50% of image size, intensity 200–230) with soft edges to simulate contamination defects. Bounding box annotations are automatically generated.

2 Intermediate Results

2.1 Augmentation Pipeline Visualization

Figure 1 shows the data augmentation pipeline used in Stage 2 training. The original PCB image is progressively transformed to simulate lithography imaging conditions.

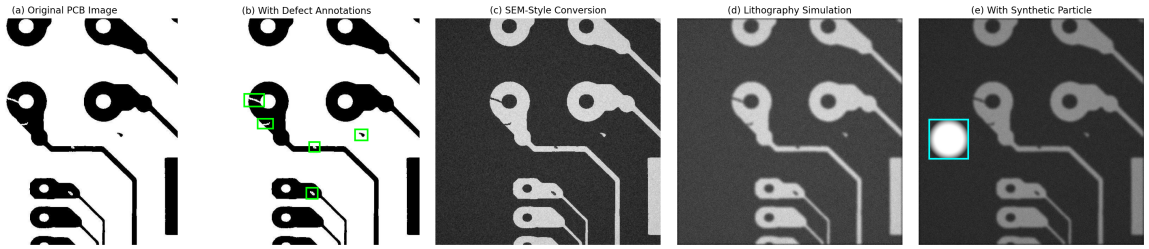
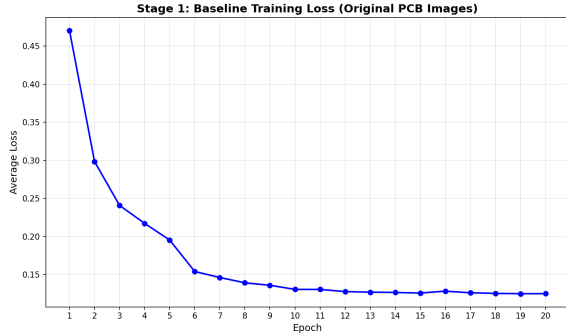


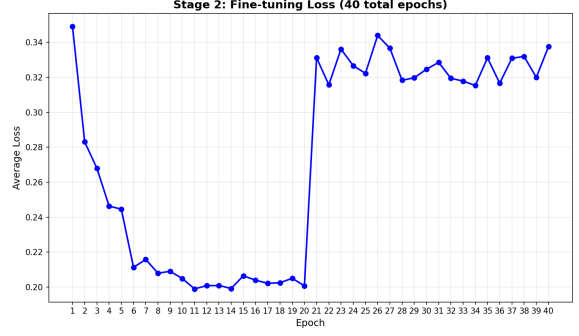
Figure 1: Data augmentation pipeline: (a) Original PCB image, (b) Original image with ground truth defect annotations (green boxes), (c) SEM-style conversion with noise and vignetting, (d) Optical lithography simulation with PSF blur and resist development, (e) Synthetic bright particle injection with auto-generated bounding box (cyan).

2.2 Training Loss Curves

Figures 2a and 2b show the training loss progression for both stages.



(a) Stage 1: Baseline training on original PCB images.



(b) Stage 2: Fine-tuning with lithography augmentation. Note: The loss spike between epochs 20–21 occurs because we updated the synthetic particle augmentation parameters (size and intensity range) at that point, requiring the model to adapt to the new augmentation distribution.

Figure 2: Training loss curves for both stages. The loss decreases steadily, indicating successful convergence.

3 Results on Test Images

The trained model was evaluated on real lithography defect images. For each test case, we present the original inspected image alongside the detection results with bounding boxes and confidence scores.

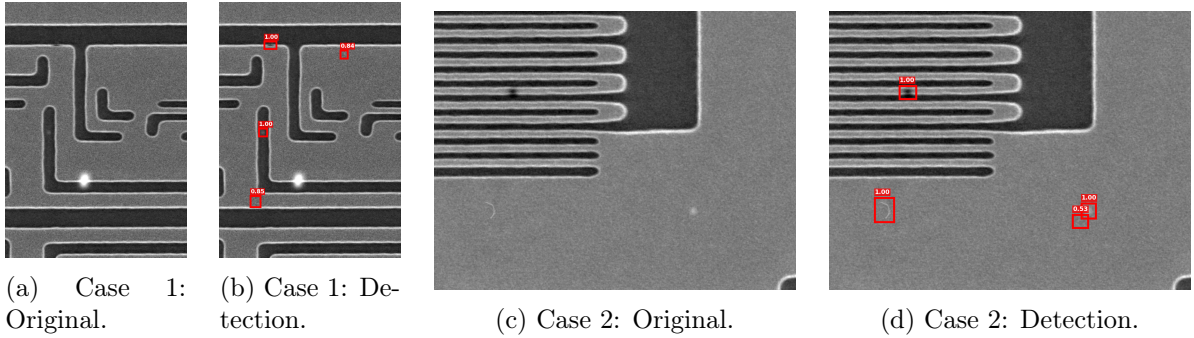


Figure 3: Defect detection results on test images. Red bounding boxes indicate detected defects with confidence scores.

4 Summary

- The two-stage training approach successfully bridges the domain gap between PCB training data and lithography target images.
- Synthetic particle injection enables detection of bright contamination defects without requiring manually annotated examples.
- The physics-based lithography simulation (Airy disk PSF, resist development) produces realistic augmented training data.
- Faster R-CNN with ResNet-50-FPN provides robust multi-scale defect detection with interpretable confidence scores.